Field	Value
Document No.	WI-001
Title	CNC Mill Setup and Operation
Revision	B
Effective Date	2026-08-10
Prepared By	Owner
Approved By	Jacob Honick, Owner
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Next Review	2027-08-10

WI-001 CNC Mill Setup and Operation

1. Scope and Confirmed Setup

Applies to CNC-MILL-001, the 3040 three-axis CNC mill with Mach3 control, in the single Torchworx LLC shop. Tool changes are manual. No probing or automatic tool-setting capability is used. The owner confirms machine, tooling, workholding and inspection capability for each job before accepting work. General industry practice is guidance, not a claim that optional equipment or features are installed.

2. Job Preparation and Safety

1. Review the current drawing, purchase order, QF-004 traveler, operation sequence and inspection plan. Confirm that every requirement can be produced and verified with owner-confirmed available methods.
2. Check machine condition and applicable maintenance under QP-019. Resolve conditions affecting safe operation or conformity before use.
3. Clean the work area and select suitable available workholding and cutting tools. Check mounting, grip, clearance and alignment against the needs of the job.
4. Wear suitable eye protection, secure loose clothing and hair, and keep hands clear of rotating tools and moving axes. Stop the spindle and motion before setup changes, tool changes, chip removal or measurement. Remove setup tools before starting. Use the machine's actual guards and safety controls; do not bypass safeguards.

3. Setup and Verification

1. Mount and align the workholding using a method selected by the owner for the required tolerances. Record significant setup requirements on the traveler or setup notes.
2. Select the first cutting tool and install it manually with the spindle stopped. Establish tool length and work datums manually using an appropriate owner-confirmed method. Enter and check the required offsets in Mach3.
3. Load and verify the correct program and revision under WI-004. Confirm that tool-change stops and manual-change instructions match the actual setup.
4. Before each manual tool change, stop motion and the spindle, install and secure the correct tool, re-establish or verify its offset, remove setup tools and check clearance before resuming.
5. Prove out new or revised programs using suitable available control functions and conservative motion settings. Do not assume an automatic changer or tool-setting cycle.
6. Produce and fully inspect the first piece under WI-005 before continuing.

4. Production and Closeout

1. Monitor cutting, tool condition, workholding and chip control. Apply cutting fluid only where suitable for the actual machine, material and setup.
2. Perform full inspection by default under QP-006 and QP-007. A different inspection extent requires a customer-specified or customer-approved written plan in the job record. Record measurements on QF-005 and QF-006 as applicable.
3. Stop for abnormal conditions or nonconformance, hold affected product and apply QP-008. Re-verify the setup and first piece after corrections before resuming.
4. Record operation completion on QF-004. Clean the machine safely, protect tooling and workholding, and record maintenance needs. Apply status tags under WI-008.

5. References

- QP-005 Production Planning and Control; QP-006 In-Process Inspection; QP-007 Final Inspection and Product Release
- QP-008 Nonconforming Product Control; QP-019 Preventive Maintenance
- QF-004 Job Traveler; QF-005 In-Process Inspection Record; QF-006 Final Inspection Report; QF-027 Equipment List
- WI-004 CNC Program Verification; WI-005 First Piece Inspection; WI-008 Part Marking

6. Revision History

Rev	Date	Description	Approved By
A	2026-08-06	Initial release	Shop Owner

Revision B Approval

Approved 2026-11-05 by Jacob Honick, Owner. Reflect the 3040 Mach3 mill, manual tool changes and manual setting; use owner-confirmed job capabilities.